A Decentralized Forecast Aggregation Market for Global Weather Prediction: Architecture and Preliminary Performance Report of Zeus Subnet 18

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***Abstract*—While data-driven weather prediction (DDWP) continues to narrow the performance gap relative to traditional numerical models, the field remains constrained by a heavy reliance on centralized supercomputing infrastructure. Zeus Subnet 18 circumvents this bottleneck by decentralizing the computational load through a global forecast aggregation protocol on the Bittensor network. This paper details the Zeus architecture, specifically its three-phase commit-reveal protocol designed to aggregate global forecasts while neutralizing adversarial strategies such as look-ahead bias and post-hoc selection. Functioning as a decentralized meta-model, Zeus incentivizes the discovery of residual accuracy through information arbitrage; participants are encouraged to synthesize diverse methodologies, including proprietary foundation models and institutional ensembles, provided all states are cryptographically anchored prior to the operational information barrier. We benchmarked the network's aggregated output on a global 0.25° grid against ERA5 reanalysis and institutional baselines (IFS HRES and AIFS) during an 18-day evaluation window. By evaluating both the internal incentive metric ($iwRMSE$) and standard scientific benchmarks ($wRMSE$), we demonstrate that the protocol effectively identifies and rewards models that optimize for specific atmospheric regimes. These results are presented as an early-stage audit of the protocol's selection and verification logic rather than a definitive claim of generalized atmospheric superiority.**

I. Introduction

Meteorology is currently undergoing a massive shift: moving away from traditional, compute-heavy numerical solvers toward fast, transformer-based AI models. Breakthroughs like GraphCast [1] and Pangu-Weather [2] have proven the efficiency of data-driven forecasting. However, the development of these systems remains largely centralized within a few institutions, which can limit the breadth of operational dissemination.

Unlike monolithic forecasting systems, Zeus does not utilize a single modeling architecture. Instead, it establishes a *decentralized meta-model architecture*,an algorithmic market where independent participants compete to discover the optimal atmospheric state under strict temporal verification constraints. Drawing on principles of forecast combination theory [3], Zeus acts as an incentive layer for information arbitrage, encouraging miners to find the optimal synthesis of institutional boundary conditions and proprietary AI foundation models to minimize realized error.

Evaluating such a decentralized network brings unique challenges, particularly in verifying that participants are not exploiting look-ahead or selection bias. To ensure scientific integrity, we rely on a strict, time-locked commit-reveal protocol. This paper details the Zeus architecture and benchmarks the network's aggregated forecasting output against the ECMWF's Integrated Forecasting System (IFS) [4] and the data-driven AIFS [5], demonstrating the capacity of decentralized incentives to surface competitive meteorological products.

II. System Architecture

Zeus operates as a continuous tournament where miners are evaluated on their ability to predict the atmospheric state across a global 0.25° grid.

A. The Three-Phase Verification Cycle

To prevent miners from exploiting historical data or adjusting forecasts mid-flight, Zeus employs a phased protocol spanning approximately 20 days.

1) Phase 1: Cryptographic Commitment: At initialization (T_0), participants must establish a verifiable commitment to their predicted atmospheric state. This process covers four primary variables: 2-metre temperature (t_{2m}), 100-metre U- and V-components of wind (u_{100}, v_{100}), and surface solar radiation downwards ($ssrd$). Each miner submits a SHA-256 hash that incorporates both the compressed prediction (P) and the competitor's unique hotkey address (A_{hk}):

$$H = \text{SHA256}(P_{\text{compressed}} \parallel A_{hk}) \tag{1}$$

This architectural choice serves a dual purpose. First, by anchoring the hash to a distributed ledger, the protocol generates a trustless, immutable timestamp. This provides a rigorous audit trail for the performance metrics reported in this study, ensuring that the forecasts were finalized before the operational information barrier.

Second, the inclusion of the hotkey address,a unique cryptographic identifier for each participant,prevents adversarial "relay mining." In a decentralized and trustless environment, participants are incentivized to plagiarize high-performing hashes. By salting the commitment with A_{hk}, the protocol ensures that identical predictions from different miners result in distinct, non-transferable hashes. This forces every participant to provide their own underlying data during the reveal phase, thereby maintaining the integrity of the competitive market and the scientific validity of the aggregated output.

2) Phase 2: Operational Data Utilization (Top-K): Once the cryptographic hashes are locked in, the validator selects a *Top-K* pool of miners (currently configured to $K = 10$) for each specific variable (t_{2m}, u_{100}, v_{100}, $ssrd$). This selection is based strictly on the miners' historical rolling ranks. We then prompt these top performers to reveal their unhashed, compressed predictions; the validator verifies these against the previously submitted hashes to ensure they match the original commitments before utilizing the data to feed the subnet's live operational stream.

To ensure continuous data availability, the system utilizes the forecast from the highest-ranked miner who successfully completes the reveal process. Because we establish this hierarchy *a priori*,before any actual forecast data is exposed,we substantially mitigate the risk of "cherry-picking" or post-hoc selection bias.

3) Phase 3: Final Scoring and Verification: Final scoring is initiated only upon the release of ERA5 reanalysis data [6]. Given the 360h (15-day) forecast horizon and the standard five-day latency of ERA5 availability, the final verification cycle occurs approximately 20 days after the initial commitment. During this phase, miners must reveal the original data corresponding to their Phase 1 hash.

The validator first verifies the cryptographic integrity of the submission by matching the revealed data against the original SHA-256 commitment. Upon a successful match, the validator calculates the forecast error and updates the miner's global rank. To ensure the transparency of the competitive market, any instance of protocol non-compliance,such as a hash mismatch or a failure to reveal data,results in a total incentive penalty for that epoch. This policy ensures that miners cannot selectively withhold poor results to protect their rolling historical ranks, thereby maintaining the scientific validity of the aggregated performance data.

B. Operational Scheduling and Latency Profile

To ensure synchronization with global meteorological standards, Zeus follows a strict temporal schedule defined by three primary variables:

- **Base Time (Initial Step):** The protocol synchronizes with the standard four-cycle synoptic schedule: 00, 06, 12, and 18 UTC.
- **Data Cut-off Window:** The maximum allowable time for miners to submit their cryptographic hashes is $T_0 + 60$ minutes. While the protocol prompts for submissions starting at 30 minutes post-initialization, the one-hour ceiling accounts for network latency and retry attempts. This ensures a robust "information barrier" while maintaining participation rates.
- **Dissemination Latency:** The live operational stream is typically published within 105 minutes of the base time. This accounts for the 60-minute commitment window followed by a transition to the data reveal phase, which commences at $T_0 + 90$ minutes. System logs indicate that the subsequent ingestion of global tensors into the operational cache (e.g., Redis), along with outlier filtering and data reconstruction, is typically completed within 7 to 15 minutes.

C. Deterministic Participant Selection

To neutralize selection bias, Zeus utilizes a *Historical Rank-Based Selection* protocol.

- **Selection Timing:** The identification of the primary data provider occurs at T_0, immediately following the hash commitment.
- **Causal Barrier** ($N = 8$)**:** Selection is based exclusively on the rolling average $iwScore$ from the preceding $N = 8$ scored epochs. This window size ensures that the system is responsive to recent model optimizations while maintaining a causal barrier against the current forecast's realized accuracy.
- **Fallback Logic:** To ensure service continuity, if the Rank 1 miner fails to provide an unhashed reveal in Phase 2, the system defaults to the subsequent highest-ranked performer.

III. Security Framework and Data Integrity

In a decentralized market, participants are financially incentivized to maximize their scores, which creates a risk of "look-ahead bias",where a miner might attempt to submit observations as predictions. Unlike centralized institutions that rely on internal trust, Zeus implements a rigorous verification framework to ensure that every forecast is a genuine forward-looking prediction rather than a retrospective data match. This section outlines the protocol safeguards designed to protect the scientific validity of the network's output.

A. Allowed Inputs and Upstream Dependency

Zeus imposes no restrictions on upstream forecasting methodology, provided forecast states are cryptographically committed before the operational cutoff time. Participants are permitted to utilize institutional forecasts, AI foundation models, proprietary ensembles, or statistical correction pipelines.

B. Forbidden Actions

To maintain the information barrier, the protocol strictly forbids:

- **Post-cutoff modification:** Altering a forecast after the T_0 temporal anchor.
- **Post-hoc selection:** Choosing an optimal model after observing ground-truth data.
- **Reveal mismatch:** Submitting a plaintext prediction that does not cryptographically match the Phase 1 commitment.

C. Market Discovery Model

Crucially, the protocol does *not* prohibit the utilization of prior-cycle institutional forecasts (e.g., T_{-6} or T_{-12} data) as boundary conditions. Instead, Zeus functions as a competitive discovery engine for the "residual accuracy" that centralized, monolithic models often fail to capture. If a participant achieves a superior $iwScore$ by applying high-speed statistical correction, bias adjustment, or neural post-processing to existing signals, the market has successfully discovered a value-added optimization. By treating miner pipelines as competitive "black boxes," the protocol incentivizes the most efficient possible use of all available atmospheric information,whether generated from first principles or refined via ensemble post-processing,thereby reducing the gap between raw institutional output and realized ground-truth states.

IV. Benchmarking Methodology

To balance what the market wants with strict scientific standards, we evaluate the network using two distinct metrics.

A. Ground Truth and Operational Baselines

To provide a rigorous evaluation of Zeus Subnet 18, we utilize a dual-layer benchmarking framework that distinguishes between retrospective ground truth and real-time institutional performance.

1) Ground Truth: ERA5 Reanalysis: The primary verification target is the **ERA5 reanalysis** dataset [6]. Produced by the ECMWF, ERA5 represents the global atmospheric state by assimilating vast quantities of historical observations into a coherent, 0.25° (≈28km) hourly grid. This dataset serves as the objective "truth" against which all forecasts are scored after the standard five-day publication latency.

2) Institutional Baselines: IFS and AIFS: We evaluate the network against two industry-standard benchmarks:

- **IFS HRES:** The ECMWF's flagship numerical weather prediction (NWP) model.
- **AIFS:** The ECMWF's state-of-the-art, data-driven AI forecasting system [5].

By comparing Zeus against both a physics-based solver (IFS) and a high-performance transformer model (AIFS), we can isolate the specific "information arbitrage" value added by the decentralized market.

3) Data Retrieval and Synoptic Constraints: Baseline data is ingested via the *Herbie* open-source library [7], which provides a unified interface for multi-cloud repositories including AWS, Google Cloud, and Azure.

To ensure maximum data integrity and coordinate parity, we impose two specific operational constraints:

1) **Synoptic Cycle Selection:** Evaluation is restricted to the **00 UTC and 12 UTC** synoptic cycles. These major cycles are selected due to the comprehensive availability of extended-horizon (360h) lead times in public cloud-native repositories, which are often truncated or unavailable for the intermediate 06 UTC and 18 UTC "minor" runs.
2) **Predictability Horizon:** While Zeus generates forecasts up to a $T+15$ day (360h) horizon, primary performance claims are focused on the **0–240h window**. This aligns with the traditional atmospheric predictability horizon; data beyond 240h is provided for protocol transparency but is treated as secondary due to the dominance of climatological noise over deterministic skill at such ranges.

This methodology ensures strict grid alignment (0.25°) across all models, eliminating potential regridding artifacts that could skew the $wRMSE$ verification.

B. Benchmark Fairness and the Temporal Information Barrier

A critical requirement for validating the decentralized aggregation model is the strict enforcement of a temporal information barrier. As detailed in Section II-B, the Zeus commitment window (Phase 1) closes no later than $T_0 + 60$ minutes. In contrast, premier institutional baselines such as the ECMWF IFS HRES and the data-driven AIFS typically enforce a dissemination delay of 5 to 6 hours post-initialization due to computational overhead and institutional release schedules.

Because the Zeus hashing period concludes hours before the concurrent T_0 institutional forecasts are made public, it is impossible for participants to "mirror" or copy the baselines against which they are evaluated. While miners may utilize prior cycle initializations (e.g., T_{-6} or T_{-12} data) as boundary conditions for their local pipelines, the forward inference for the current epoch must be generated independently to meet the commitment deadline.

Consequently, the Zeus operational stream represents a genuine forward-predictive aggregation. By the time the concurrent institutional benchmarks become available for comparison, the Zeus forecasts have already been cryptographically locked, time-stamped, and,in many cases,disseminated to the operational stream. This temporal separation ensures that any observed skill improvements are a product of decentralized model diversity rather than data leakage.

C. Metrics: Scientific Audit vs. Operational Incentive

Zeus employs a dual-metric evaluation framework to bridge the gap between global scientific accuracy and localized economic utility. This architectural choice enables a neutral benchmark of the network's performance against institutional baselines while concurrently steering the decentralized market via a specialized incentive layer. In this preliminary proof-of-concept (PoC) phase, the reward structure is intentionally calibrated as a strategic middle ground. By diversifying the scoring criteria, the protocol prevents participants from over-optimizing for the specific statistical biases inherent in any single error metric,a critical safeguard in a decentralized market where miners are primarily incentivized by score maximization rather than generalized atmospheric skill.

1) Scientific Audit: $wRMSE$: For our scientific audit, we use the standard latitude-weighted RMSE ($wRMSE$). This metric strips away all geographic scalars to provide a neutral, apples-to-apples evaluation:

$$wRMSE = \sqrt{\sum_{i=1}^{n} w_i(\tilde{X}_i - X_i)^2}, \quad w_i = \frac{\cos(\phi_i)}{\langle\cos(\phi)\rangle} \tag{2}$$

The spatial weights w_i compensate for the varying density of grid points near the poles. Using this standard metric ensures that the Zeus "Best Miner" is evaluated on the exact same global scale as institutional benchmarks like GraphCast [1] and AIFS [5].

2) Operational Incentive: $iwScore$: While $wRMSE$ measures global skill, the $iwScore$ drives the network's economy. It dictates participant selection and determines miner rewards by blending the incentivized-weighted (iw) variants of RMSE and Mean Absolute Error (MAE):

$$\text{iwScore} = 0.5 \cdot \text{iwRMSE} + 0.5 \cdot \text{iwMAE} \tag{3}$$

The "i" (incentivized) component applies area scalars (S_{area}) to steer miner compute toward high-value geographic zones. Instead of treating the entire globe equally, these scalars force miners to optimize for regions with heavy economic demand,such as the European energy corridor and Central German grid hubs (Fig. 1). The constituent metrics are defined as:

$$iwRMSE = \sqrt{\frac{\sum_i S_{area,i} w_i (\tilde{X}_i - X_i)^2}{\sum_i S_{area,i} w_i}} \tag{4}$$

$$iwMAE = \frac{\sum_i S_{area,i} w_i |\tilde{X}_i - X_i|}{\sum_i S_{area,i} w_i} \tag{5}$$

By blending a squared-error term (RMSE) with a linear-error term (MAE), the $iwScore$ incentivizes miners to minimize average error magnitude while strictly punishing large "forecast busts" that could compromise regional energy-grid stability.

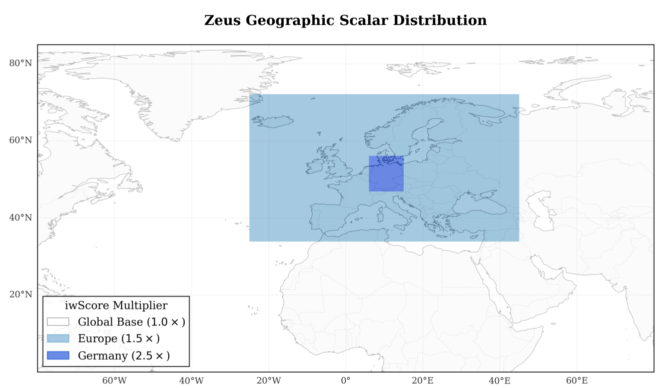

Fig. 1. Spatial distribution of geographic scalars S_{area} applied to the global 0.25° grid to prioritize high-precision forecasting in regions of significant economic utility.

V. Experimental Framework and Operational Constraints

To establish a rigorous comparative environment, Zeus Subnet 18 enforces strict temporal and spatial constraints. These parameters are designed to prevent data leakage and provide a transparent framework for measuring operational latency.

A. Data Encoding and Reconstruction Fidelity

Moving global atmospheric tensors across a decentralized network requires balancing bandwidth constraints with scientific precision. To achieve this, Zeus utilizes a half-precision quantization and entropy encoding pipeline designed for high-throughput validation.

- **Quantization:** Global tensors are cast to 16-bit half-precision (FP16). While this is a step down from standard FP32, it preserves the necessary dynamic range for meteorological variables while reducing the raw tensor memory footprint by approximately 50%.
- **Compression Pipeline:** The protocol utilizes the Blosc2 framework [8] with a Zstandard (ZSTD) codec [9] and Bitshuffle filtering. For a global 0.25° grid (1440×721 points) across a 360h hourly horizon, the raw FP16 payload per variable is approximately 750 MB. In practice, the Zstandard codec achieves compression ratios that reduce the operational payload to ≈180–250 MB per miner.
- **Throughput Feasibility:** With a typical competition density of 10 miners per variable, a validator ingests ≈2–2.5 GB of compressed data per epoch. Given the incentivized nature of the validator tier, which mandates professional-grade network infrastructure (e.g., 1 Gbps symmetrical), this total volume is ingested in under 30 seconds, well within the 10-minute operational window observed in system logs.

B. Spatial Fidelity and Grid Enforcement

Rather than allowing miners to rely on arbitrary interpolation or latent-space forecasting grids, Zeus enforces a strict 0.25° (≈28km) global coordinate system.

- **Standardized Input/Output:** Miners must output a 3D vector explicitly mapping latitude, longitude, and time steps.
- **Validation:** Submissions that do not strictly conform to this coordinate system are automatically rejected. This eliminates regridding artifacts at the validation layer and ensures the $iwScore$ is calculated on raw, high-fidelity miner output.

C. Operational Latency and the Data Assimilation Tradeoff

A primary operational characteristic of the Zeus architecture is its dissemination speed. Institutional models like the ECMWF IFS HRES typically face a 6-hour delay between initialization and dissemination. This delay is largely driven by intensive data assimilation (DA) workflows, which require waiting for a critical mass of global observations to be recorded and processed into a precise starting state.

In contrast, Zeus aggregates forecasts approximately 1.5 hours post-initialization. This latency reduction represents a fundamental architectural tradeoff: Zeus miners largely utilize boundary conditions from prior institutional cycles (e.g., T_{-6}) or rapid AI-inference models rather than performing real-time 4D-Var data assimilation of raw observations. While this bypasses the primary bottleneck of centralized NWP, it introduces a necessary reliance on the quality of existing upstream signals.

This comparison should not be interpreted as evidence that decentralized hardware is computationally superior to centralized supercomputing infrastructure. Rather, it highlights that Zeus serves as a high-speed optimization and dissemination layer, prioritizing reduced publication latency over the observation-heavy initialization workflows typical of institutional centers.

VI. Performance Results and Discussion

We evaluated the Zeus operational data stream during a specific 18-day test period from April 3 to April 21, 2026. Because the atmosphere was in a state of relative synoptic equilibrium during this period, it provides a clean baseline to assess standard model skill. Consistent with our deterministic selection rules, the results below represent the output of the primary data provider, dynamically selected using the rolling $iwScore$.

A. Global Deterministic Skill and Statistical Rigor

We report the area-weighted Root Mean Square Error ($wRMSE$) for 2m temperature (t_{2m}), 100m wind components (u_{100}, v_{100}), and surface solar radiation downwards ($ssrd$) recorded during the April 3–21, 2026 audit. All forecasts are benchmarked against the ERA5 global reanalysis at a 0.25° (≈28km) resolution. To ensure a conservative statistical framework and avoid pseudo-replication across spatial dimensions, we collapsed the global grid points into a single latitude-weighted $wRMSE$ scalar per lead-time step prior to performing comparative analysis.

TABLE I
AGGREGATED 0–240H wRMSE AND OBSERVED RELATIVE IMPROVEMENT

Variable	Zeus wRMSE	vs. IFS HRES	vs. AIFS
2m Temp. (t_{2m}) [K]	1.7775	+23.55%	+10.90%
100m U-Wind [m/s]	3.8662	+14.17%	+6.34%
100m V-Wind [m/s]	3.9796	+14.69%	+6.63%
SSRD [J/m^2]	1.22×10^6	+16.61%	+1.75%

The observed 23.55% improvement in t_{2m} over the numerical IFS HRES (Table I) is notable but must be contextualized within the atmospheric conditions of the audit period. In traditional meteorology, deterministic gains of this magnitude typically indicate that the aggregator has successfully corrected for specific model biases prevalent during a period of low atmospheric volatility. We applied a paired t-test ($\alpha = 0.05$) across the $n = 18$ initialization cycles to evaluate the consistency of this performance delta. However, we acknowledge that the high temporal autocorrelation of synoptic weather structures reduces the effective degrees of freedom below the nominal sample size.

Consequently, while the t-test indicates point-wise statistical significance across much of the horizon, we prioritize the 95% confidence intervals (Fig. 2) as a descriptive measure of trend stability rather than a definitive p-value guarantee. This approach accounts for the inherent temporal dependence of the audit period while demonstrating the robustness of the decentralized selection mechanism. These results should be viewed as a validation of the Zeus protocol's ability to identify the "optimal" local signal within a specific regime, rather than a claim of generalized superiority over institutional models across all seasonal extremes.

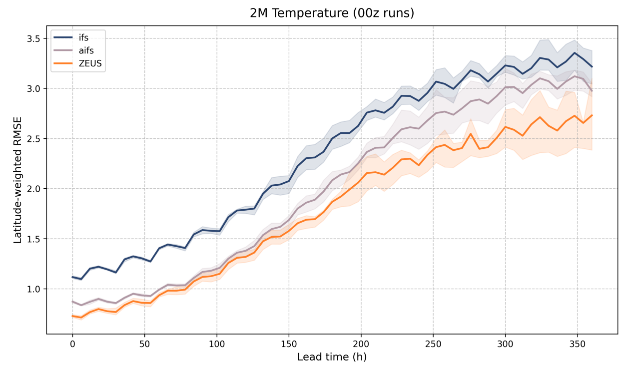

Fig. 2. Global wRMSE for 2m Temperature (April 3–21, 2026) against ERA5 ground truth, featuring 95% confidence intervals over the $n = 18$ initialization cycles.

B. Analysis of Operational Gains and Structural Affinity

The reported 23.55% improvement in t_{2m} over IFS HRES represents an extraordinary deterministic gain that requires careful diagnosis to ensure scientific validity. A primary factor in this delta is the structural relationship between the ERA5 ground truth and the IFS-family baselines. Because ERA5 is produced using a 4D-Var data assimilation system closely related to the IFS, there exists an inherent statistical affinity between the reanalysis target and the institutional products utilized by miners.

In a decentralized market, miners are incentivized to optimize specifically for the verification target (ERA5). We hypothesize that the network's top performers are successfully executing high-speed "regression toward the reanalysis",effectively post-processing institutional boundary conditions to align more closely with the specific statistical distribution of ERA5. These results should therefore be interpreted as a validation of the Zeus selection mechanism: it demonstrates that an incentivized market can identify and reward the precise "information arbitrage" required to correct persistent institutional biases relative to a specific target reanalysis. This "proof-of-life" for the protocol suggests that decentralized aggregation can surface superior operational utility even when constrained by the predictability limits of the underlying atmospheric state.

C. Wind Vector Performance and Convergence

For 100m wind components, Zeus maintains a consistent lead over the numerical IFS HRES baseline (+14.17% for u_{100}). When compared to AIFS, the models show competitive parity in the 72–120h range, where the performance differences fell within the estimated confidence intervals. However, Zeus regains a statistically significant lead in the long-range horizon ($>$ 240h), suggesting the decentralized aggregation market effectively mitigates the model drift inherent in extended deterministic lead times (Fig. 3).

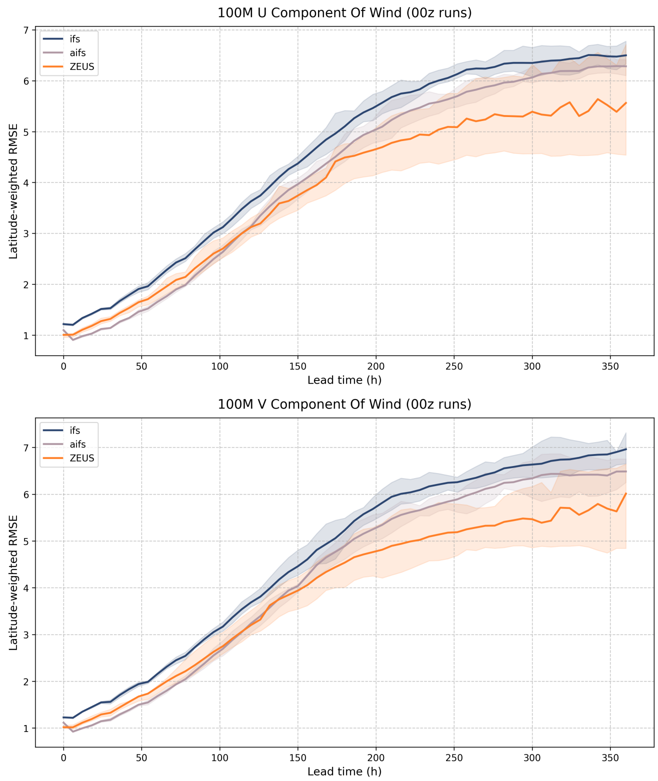

Fig. 3. wRMSE comparison for 100m U (left) and V (right) wind components across the 360h horizon.

D. Dynamic Model Agility and Adaptive Selection

A primary differentiator of the Zeus architecture is its ability to treat participant methodologies as proprietary, competitive "black boxes." During the April audit, the network dynamically pivoted between independent miner architectures as relative skill fluctuated across different weather regimes. By surfacing variable-specific specialists for t_{2m} versus u_{100}, the protocol creates a highly adaptive operational stream. This market-driven selection allows the output to remain "methodology-agnostic," ensuring that whether miners utilize independent foundation models or sophisticated ensemble pipelines, the most accurate atmospheric state is prioritized. This mechanism helps reduce persistent model-specific forecast drift by continuously reallocating operational weight toward recently successful pipelines that have navigated the economic pressure of the tournament.

E. Surface Solar Radiation and Hourly Utility

Zeus exhibits an overall improvement of 16.61% against IFS for $ssrd$. While the improvement against AIFS is a more modest 1.75% (Fig. 4), the Zeus system provides native hourly resolution. This is a critical operational advantage over the 6-hourly intervals provided by standard institutional models, as it captures the intra-window solar irradiance fluctuations that energy grid managers require.

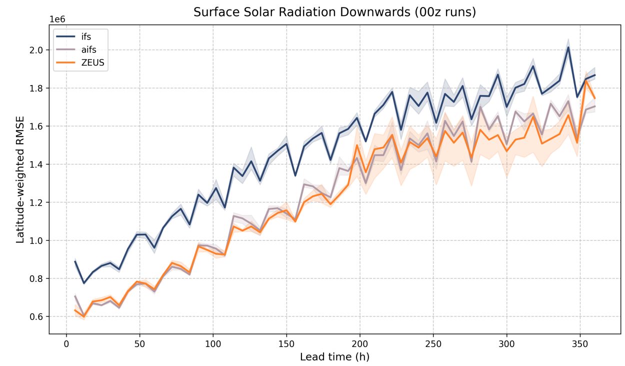

Fig. 4. wRMSE of SSRD accumulated forecasts over the 360h horizon.

F. Targeted Incentive Density: The European Scalar

During the evaluation period, we applied geographic scalars (S_{area}) to prioritize the European sector, specifically Germany.

- **Incentive Alignment:** While we use global $wRMSE$ for scientific reporting, miner optimization is strictly driven by the $iwScore$, directing computational resources toward these high-density energy hubs.
- **Operational Impact:** The sharp performance in $ssrd$ and wind components across these sectors demonstrates that the network can provide high-fidelity localized forecasting competitive with operational regional forecasting requirements for energy-oriented applications.

G. Dissemination Advantage and Extended Horizon

Zeus forecasts extend to a $T + 15$ day (360h) horizon, maintaining architectural stability well beyond the standard 240h limit of many operational models. Furthermore, the 1.5-hour dissemination time represents a ≈75% reduction relative to the standard 6-hour delay of institutional NWP models. These results suggest that decentralized forecast aggregation systems may offer substantial dissemination-speed advantages relative to centralized operational workflows.

H. Discussion: Significance of Early-Stage Observation

Despite the relatively brief evaluation window, the consistency of the observed improvements across initialization cycles suggests the results are unlikely to be explained solely by short-term seasonal variability. However, broader validation across multiple atmospheric regimes and transition seasons remains necessary before drawing strong conclusions regarding long-term, generalized forecasting skill. Accordingly, Zeus should be interpreted primarily as a decentralized forecast aggregation and selection architecture rather than a fully independent atmospheric foundation model.

VII. Limitations

While the Zeus architecture demonstrates significant operational utility, we must acknowledge several constraints in our current evaluation:

- **Evaluation Window and Regime Bias:** Our study spans a relatively short, 18-day window in April 2026 characterized by general synoptic equilibrium. This lacks the broad seasonal validation required to prove robustness during rapid regime shifts or extreme weather events.
- **Upstream Dependency and Arbitrage:** While miners are permitted to ingest external products, the network's identity is that of an aggregator and optimizer. This creates an inherent link to the broader forecasting ecosystem; the network's performance floor is tied to the quality of public institutional data, even as its performance ceiling is defined by miner-driven optimizations.
- **Methodological Opacity:** The decentralized nature of the tournament treats participant pipelines as proprietary "black boxes." While this protects the intellectual property of miners and maximizes competitive pressure to produce accurate data, it limits the scientific interpretability of the specific atmospheric physics or neural architectures driving a winning forecast.
- **Metric Weight Sensitivity:** The equal weighting of $iwRMSE$ and $iwMAE$ in the $iwScore$ serves as a strategic baseline for this proof-of-concept to prevent miners from over-fitting to a single statistical moment. However, this 50/50 split has not yet undergone formal sensitivity analysis to determine the optimal balance for specific industrial use cases, such as energy grid management where the cost of outliers may outweigh mean error.
- **Lack of Ablation Studies:** Future study is required to fully isolate the independent performance contributions of the Top-K routing versus geographic scalars across all variables, which will help us better understand the individual impact of the network's constituent mechanisms.
- **Limited Probabilistic Evaluation:** This preliminary audit focuses exclusively on deterministic metrics (e.g., $wRMSE$). A comprehensive probabilistic evaluation is necessary to fully assess the network's reliability and ensemble spread.
- **Incentive Bias:** Geographic scalars deliberately skew optimization toward specific economic zones, meaning global scientific accuracy may occasionally take a backseat to regional market demands.

VIII. Reproducibility and Data Provenance

Zeus Subnet 18 maintains a policy of open-access data to facilitate independent verification of the results presented in this audit. Global forecast tensors and corresponding metadata are hosted at: https://data.zeussubnet.com/.

A. Evolution of the Trustless Verification Layer

A core objective of this work is the establishment of a verifiable, trustless audit trail for meteorological data. During the formalization of this paper, it became evident that while internal validator logs maintained the integrity of the commitment cycle, a higher level of decentralization was required for external scrutiny. This realization catalyzed a protocol enhancement: the direct anchoring of SHA-256 hashes to the Bittensor blockchain.

Consequently, while the performance metrics for the specific 18-day window analyzed in this study rely on validator-side logs and Parquet-level metadata, the protocol has since transitioned to a fully trustless model. All subsequent and future forecast cycles are anchored to immutable block headers. This ensures that the "information barrier" is now enforced by a distributed ledger, allowing any third party to verify the temporal provenance of the predictions without relying on centralized attestation.

B. Scope of Preliminary Findings

It is important to reiterate that this paper characterizes the *preliminary operational signs* of the Zeus architecture rather than providing a generalized atmospheric baseline. The 18-day audit period serves as a proof-of-concept for the subnet's selection and aggregation mechanisms during a window of relative synoptic stability.

To ensure longitudinal transparency, the project will continue to upload all operational data to the dedicated repository. This ongoing dissemination will allow the scientific community to observe how the protocol performs across broader seasonal transitions and high-volatility regimes, building upon the promising initial results documented in this report.

IX. Conclusion and Future Work

Zeus Subnet 18 demonstrates that decentralized forecast aggregation markets are operationally viable. By enforcing strict cryptographic timing verification, the protocol proves that an adaptive, competitive selection mechanism can surface verified operational forecasts while neutralizing look-ahead bias. The network indicates that decentralized pipelines can offer latency advantages by optimizing existing signals, albeit with a recognized tradeoff regarding real-time data assimilation workflows typical of institutional modeling centers.

We acknowledge that the 18-day evaluation period presented in this audit represents an early-stage observation of a nascent protocol. Because Zeus Subnet 18 is a live, decentralized market utilizing immutable cryptographic commitments, its historical performance cannot be retroactively manufactured or extended beyond its operational inception. Consequently, while the reported improvements are statistically significant for the audit period, they should be interpreted as a validation of the protocol's optimization and selection logic during a window of low volatility. The 23.55% lead in t_{2m} demonstrates that the market can successfully "fine-tune" atmospheric states for specific regimes, though long-term performance may regress toward standard scientific margins as the network encounters higher atmospheric variance. The primary contribution of this work is the validation of a decentralized incentive structure that can consistently surface and verify forecast products at a speed that traditional centralized systems,which prioritize observation-heavy initialization workflows,currently do not target.

Future work will focus on expanding our temporal audit window to evaluate how this adaptive selection mechanism performs under extreme atmospheric volatility and varied seasonal transitions. Additionally, we plan to conduct broader ablation studies on the decentralized routing mechanisms and expand our metric framework to include probabilistic evaluation, further testing the resilience of decentralized economic incentives against complex meteorological anomalies. As the network matures, Zeus aims to move from a proof-of-concept aggregation layer to a primary global hub for decentralized meteorological data dissemination.

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